

1×20=20

A. Answer the following question with select the correct options.

1. This is the largest phylum of Animal on the earth.

- A. Mollusca B. Amphibia ☒ C. Arthropoda D. Aves

2. Where is DNA present in the eukaryotic cells?

- ☒ A. Inside the nucleus B. With other cellular contents C. Inside the ribosomes D. Not present

3. Which of the following nitrogenous base is not present in DNA?

- A. Thymine B. Adenine C. Guanine ☒ D. Uracil

4. Which of the following is the function of lysosomes?

- A. Autophagy B. Autolysis ☒ C. Digestion D. All of the above

5. Who is known as the "Father of Genetics"?

- A. Morgan ☒ B. Mendel C. Watson D. Bateson

6. Glycolysis is the conversion of

- A. Fructose into phosphoenolpyruvate B. Fructose into pyruvate
C. Glucose into phosphoenolpyruvate ☒ D. Glucose into pyruvate

7. Which substrate is used in the last step of glycolysis?

- A. Glyceraldehyde 3-phosphate B. Pyruvate
☒ C. Phosphoenolpyruvate D. 1, 3-bisphosphoglycerate

8. Which of the following immunity is present from our birth?

- ☒ A. Innate Immunity B. Active immunity C. Passive immunity D. Acquired immunity

9. Which of the following cells is involved in humoral immunity?

- A. T-cells ☒ B. B-cells C. Mast cells D. Both T and B cells

10. Anticodon is present in

- A. DNA B. tRNA ☒ C. rRNA D. mRNA

11. Name the RNA molecules which is used to carry genetic information copied from DNA?

- A. tRNA ☒ B. mRNA C. rRNA D. snRNA

12. The percentage of human genome which encodes proteins is approximately

- ☒ A. Less than 2% B. 5% C. 25% D. 99%

13. The jawless vertebrate is

- A. Crocodile B. zoris C. Hyla ☒ D. Petromyzon

14. Phylum of *Taenia Solium* is

- A. Aschelminthes B. Annelids ☒ C. platyhelminths D. Mollusca

15. On the Origin of Species was written by _____

- ☒ A. Charles Darwin B. Ludmila Kuprianova C. Mikhail A. Fedonkin D. None of the above

16. Which of these is a characteristic of prokaryotic cells?

- ☒ A. Absence of cell organelles
☒ B. Absence of nucleus
☐ C. Presence of 70S ribosomes
☒ D. All of these

17. From which structure is a mesosome derived from?

- ☐ A. Plasmid ☐ B. Cell wall ☒ C. Ribosome ☐ D. Cell membrane

18. The scaleless vertebrate is

- ☐ A. Snake ☐ B. Rohu ☐ C. Shark ☒ D. rat

19. Which of the following is a flightless bird?

- ☒ A. Pigeon ☐ B. vulture ☐ C. Parrot ☐ D. ostrich

20. Book lungs are respiratory organs in

- ☒ A. Insects ☐ B. Arachnids ☐ C. Molluscs ☐ D. Echinoderms

B. Answer any 10 questions from the following.

2×10=20

1. What is Coelom? Where is it found?
2. What is genetic code?
3. In which cell organelle thylakoid is found? Mention its one function.
4. Mention two differences between DNA and RNA.
5. Write down the two characteristic features of Phylum Arthropoda.
6. Mention the chemical reaction of photosynthesis?
7. What is Central Dogma in molecular biology?
8. Mention two differences between B-cell and T-cell.
9. What is mRNA? Mention its one function.
10. What is Green gland? Where it is found?
11. What do you mean by antibody?
12. What is chromosome?

C. Answer any 6 questions from the following. 5×6=30

1. Write down the characteristics features of Phylum Mollusca. Mention two animal names under this phylum.
2. Write down a short note on Antibody structure.
3. Describe the Oparin and Haldane theory on the chemical basis of the origin of life.
4. What is nucleotide? Write down the structure of double helix DNA.
5. What is notochord? Write down the characteristics features of class Amphibia.
6. Write down the structure of mitochondria and mention its function.
7. What is immunity? Mention the different types of WBC. Mention its function in immunity.
8. What is metazoan? Write down the characteristics features of Phylum Poifera.

D. Answer any one question from the following

10×1=10

1. What is Photosynthesis? Write down the process of Photosynthesis. Mention its importance. (2+6+2)
2. What is TCA cycle. Describe the Krebs cycle. Mention its importance. (2+6+2)